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Joint Evidence Gating for Continual Maintenance of Predictive Digital Twin Models Under Unreliable Monitoring

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ABSTRACT: Predictive digital twin models require maintenance when operating conditions change, but unreliable monitoring can make a fitted candidate worse than the deployed model. This study examines the commitment decision separately from candidate adaptation. We propose a Joint Gate that requires a negative current Verification mean loss difference and combines that difference with scaled one-sided Mann–Whitney evidence. The score is a frozen empirical decision rule, not a new statistical test or a formal safety guarantee. Candidate and parent identities and the deployment action are sealed before future Service outcomes become available. A main experiment compares twelve methods across 32 synthetic computing centers, three model seeds, and three monitoring streams, comprising 3,456 lifecycles. Joint Gate achieves clean/mixed balanced mean squared error of 0.00318977 against 0.00335236 for the selected strongest empirically risk-qualified active baseline, DualMean: a 4.85% reduction, with improvement in 29 of 32 centers. Observed harmful-commit frequencies are 1.30%, 3.65%, and 3.99% in clean, mixed, and coherent-sticky streams. Unrestricted adaptation methods attain lower raw prediction error but fail the source risk qualification, and the utility contrast with RecentMean is inconclusive. Supplementary monitoring, architecture, external-workload, and thermal-task studies delimit this result: severe mixed corruption and longer horizons exceed the source risk criterion; Joint Gate meets empirical risk criteria on exposed Alibaba machines, but its utility advantage over the fixed DualMean comparator does not transfer; and transferred source risk qualification fails in all tested thermal streams. These findings support a bounded empirical deployment method and auditable evaluation procedure, rather than domain-independent superiority or safety.

KEYWORDS: Predictive digital twin; continual maintenance; selective deployment; verification; monitoring faults; empirical risk

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1 Introduction

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Digital twin (DT) models provide synchronized virtual representations for monitoring, prediction, and control in industrial, wireless, and networked computing systems [Original reference 1, 2, 3, 4, 5]. After deployment, these models must remain aligned with operating conditions that evolve over time. Newly accumulated monitoring data enable maintenance, but they may reflect legitimate workload and resource changes or unreliable observations caused by measurement noise, stale data, communication faults, or manipulation. Directly updating the deployed model can therefore improve adaptation while also replacing a stable state with a degraded one. Figure Original 1 illustrates this maintenance setting.

A predictive digital twin (DT) maintains a model of an evolving physical or computational asset. Its predictions can become inaccurate after changes in workload, operating regime, or instrumentation [1,2]. Digital twins therefore require maintenance beyond initial model fitting [3–5]. Updating a model whenever observations arrive, however, couples two different decisions: whether a candidate can be fitted and whether that candidate should replace the deployed state. A candidate fitted to a distorted monitoring interval can propagate an error into subsequent maintenance stages [6]. Conversely, retaining the initial model indefinitely can sacrifice adaptation to legitimate change.

This paper studies deployment decisions for asset-specific forecasting models with unreliable monitoring. At each maintenance opportunity, a candidate is adapted from the deployed parent on Update data. Both models are compared on later, currently available Verification data. The deployment action is fixed before the next Service interval is accessible, and Service supplies retrospective outcome labels. The challenge is to assess a candidate when both Update and Verification observations may be distorted.

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Existing DT updating and synchronization methods maintain physical–virtual consistency as system conditions change [Original reference 6, 7, 8], while broader network-DT studies emphasize adaptive modeling and lifecycle synchronization [Original reference 9, 10]. Continual-learning methods address sequential adaptation and knowledge retention, and reliability-oriented methods detect or mitigate unreliable observations before model modification. However, these lines of work do not explicitly treat post-adaptation state replacement as a separate deployment decision. When benign operating changes and unreliable observations coexist, successful adaptation on available maintenance data does not guarantee a beneficial deployed-state transition. The remaining challenge is therefore to make candidate-to-deployed state transitions reliable while retaining useful adaptation.

Continual learning addresses adaptation and forgetting [7–9], while validation-based updating offers a natural deployment filter. The question is how the magnitude of a current loss improvement should interact with evidence about the current loss distributions. A hard conjunction can reject a substantial mean improvement when a separate rank threshold is narrowly missed. A mean threshold alone does not distinguish improvements with different distributional support. Historical constraints address retention of earlier regimes and may restrict transitions after legitimate drift.

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To address this challenge, we propose RAVM-DT, a verification-guided continual maintenance algorithm for deployed DT models. At each maintenance stage, RAVM-DT adapts the current model using update data to form a candidate, compares that candidate with the deployed state using paired recent and bounded historical verification evidence, and commits it only when both evidence criteria support replacement. Bounding the historical evidence horizon limits the influence of obsolete operating regimes while preserving temporally distinct evidence against harmful replacement.

We propose Joint Gate, a deployment rule that requires a negative current candidate-minus-parent mean loss difference and combines it with scaled one-sided Mann–Whitney evidence. The rule allows the magnitude of improvement and rank evidence to contribute jointly to the commitment decision. We

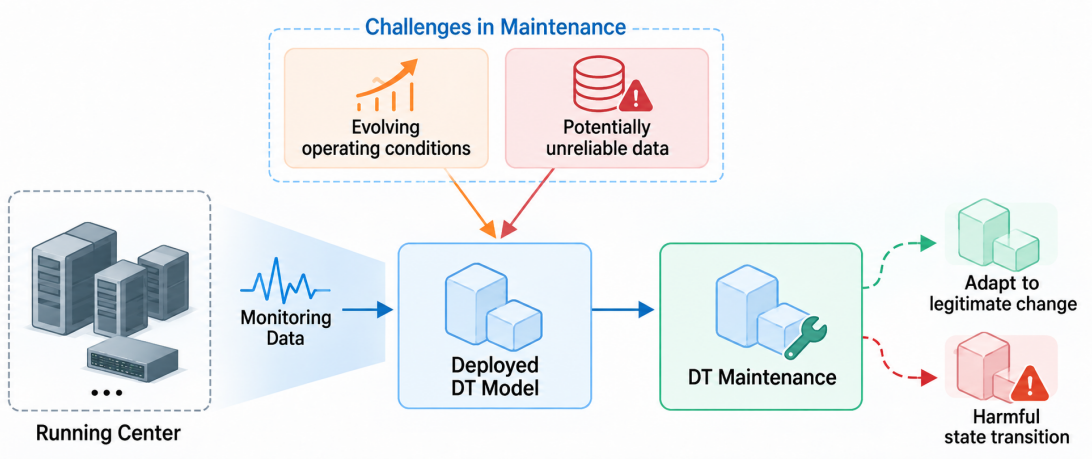


Figure 1: Predictive digital twin (DT) maintenance under changing conditions and unreliable monitoring. Harm denotes increased predictive loss.

evaluate prediction accuracy among empirically risk-qualified active methods and examine how the trade-off changes with monitoring faults, architectures, forecast horizons and prediction tasks.

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The main contributions are as follows.

- **State-transition formulation for continual DT maintenance.** We formulate maintenance under evolving and potentially unreliable monitoring data as repeated candidate-to-deployed state transitions, characterized by lifecycle loss, harmful-transition frequency, and harmful-transition downside.
- **Verification-guided selective state transition.** RAVM-DT separates candidate adaptation from deployment commitment and uses paired recent and bounded historical evidence to decide whether to commit or retain the current state.
- **Evaluation of the safety–utility–plasticity trade-off.** We evaluate the fixed RAVM-DT configuration under benign drift, multiple corruption patterns and severities, mechanism variants, model architectures, prediction horizons, maintenance cost, and an external production workload trace.

The study makes three contributions. First, it specifies a continuous mean–rank deployment score with an independent mean-improvement requirement and a causal model-state interface. Second, it compares the rule with eleven alternatives under explicit empirical risk qualifications and center-level uncertainty. Third, monitoring-stress, architecture, external-workload and measured thermal-task evaluations identify the conditions under which the rule remains useful and those in which qualification or relative utility fails.

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The remainder of this paper is organized as follows. Section Original 2 reviews related work. Section Original 3 defines the problem, Section Original 4 presents RAVM-DT, Section Original 5 reports the evaluation, and Section Original 6 concludes the paper.

Figure 1 summarizes the maintenance setting.

2 Related Work

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Original 2.1: Digital Twin Model Updating and Synchronization

2.1 Predictive Model Maintenance and Continual Learning

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Digital-twin research has increasingly extended from one-time construction to post-deployment synchronization and updating. Li *et al.* integrated continual learning into DT-enabled edge service provisioning [Original reference 6], Rodriguez *et al.* addressed data-quality control before automatic DT updating [Original reference 7], and Amad *et al.* explored continual updating with in-context large-language-model adaptation [Original reference 8]. Recent DT-network surveys likewise identify adaptive modeling, synchronization, and AI-assisted lifecycle management as central challenges [Original reference 9, 10]. These studies motivate continual maintenance, but primarily focus on how a model is updated or synchronized rather than whether an adapted candidate should replace the deployed state.

Digital twin research covers representations, synchronization and application-specific services [3,10, 11]; model maintenance is one component of this system [12–14]. Concept drift changes the generating process [15], whereas a monitoring fault changes the evidence available to the learner without necessarily reflecting the asset's evolution. This distinction motivates evaluating adaptation to legitimate change separately from responses to unreliable observations.

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Original 2.2: Continual Adaptation under Evolving Conditions
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Continual learning studies sequential adaptation and the stability–plasticity trade-off [Original reference 11, 12, 13, 14], closely related to adaptation under concept drift [Original reference 15]. Representative approaches preserve prior knowledge through parameter regularization [Original reference 16], distillation [Original reference 17], exemplar or episodic memory [Original reference 18, 19, 20], and importance-aware constraints [Original reference 21]. Recent work has also analyzed forgetting and adaptation in continual regression settings [Original reference 22], while contaminated-stream methods explicitly address unreliable sequential data [Original reference 23]. These methods improve adaptation, but an adapted model may still be unsuitable for deployment if the state transition itself is insufficiently supported.

Continual-learning techniques seek to retain useful knowledge while adapting [7,8,16]. Elastic weight consolidation (EWC) penalizes changes to parameters important to earlier tasks [17]; experience replay reuses previous observations [18]. Robust losses reduce the influence of unusual residuals [19]. These techniques modify candidate generation. A deployment gate instead determines whether a fitted candidate becomes the next parent, so adaptation and commitment are complementary interventions.

Published update mechanisms differ in their decision objects and information. ADWIN compares means in older and newer portions of an adaptive data window and removes stale observations after detecting a change [20]. It uses temporal history to revise a learner or its statistics, rather than comparing a fitted candidate with an incumbent. Its analysis assumes independent bounded observations. Seldonian regression separates candidate selection from a held-out constraint test and returns a solution only when confidence bounds support the specified behavioral constraints [21]. Its guarantees depend on the sampling assumptions and constraint estimators. Joint Gate instead compares current candidate and parent losses in a temporally dependent, potentially corrupted Verification interval; its risk performance is assessed empirically.

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Original 2.3: Reliable and Trustworthy DT Updating

2.2 Validation and Selective Deployment

Validation-based deployment compares a fitted candidate with an incumbent before replacement. The protocol controls defined in this study isolate nearby decision rules: RecentMean applies a current mean threshold; MWU-Mean adds a hard rank threshold; DualMean additionally requires a nonpositive historical mean difference. These are study-defined controls, distinct from the published ADWIN and Seldonian methods. The original RAVM rule combines robust current and bounded historical evidence.

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Reliable DT updating commonly emphasizes data-quality assessment, anomaly handling, and robust learning. Shen *et al.* evaluate anomalous physical-twin data before trustworthy power-system DT updates [Original reference 24]. Robust estimation limits the influence of outlying observations [Original reference 25], while Byzantine-robust learning and poisoning research study unreliable or adversarial training information [Original reference 26, 27]. These directions improve the reliability of data or learning processes, but our focus is complementary: after a candidate has been adapted, should it replace the current deployed state when recent operating changes and potentially unreliable maintenance evidence coexist? RAVM-DT therefore targets the reliability of post-adaptation model-state transitions rather than adaptation alone.

The Mann–Whitney procedure provides a classical rank-based comparison [22]. Joint Gate uses its one-sided output as a score feature alongside the mean loss difference. Overlapping forecasting windows and temporal dependence preclude interpreting that output as a calibrated future-harm probability. The lifecycle evaluation measures the resulting risk–utility trade-off, while a same-state replay compares decision behavior on identical inputs.

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Original 3: System Model and Problem Formulation

3 Problem Formulation and Information Boundaries

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Original 3.1: System Setting and Deployed DT Model

3.1 Roles, Candidate Adaptation, and Deployment State

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We consider a DT platform that maintains center-specific models for a set of heterogeneous computing centers C. For center $c \in C$, let a_c denote the architecture of its deployed DT model and let $\theta_{c,t}$ denote the model parameters after maintenance stage t. The deployed DT model state is represented as

DELETED Original Equation 1

$$M_{c,t} = (a_c, \theta_{c,t}), \qquad t = 0, \dots, T,$$

where T is the number of maintenance stages.

For asset c, seed r, and stage t, let f_{t-1} denote the deployed parent. The candidate $\widetilde{f}_t = \mathcal{A}(f_{t-1}, U_t)$ is fitted on Update data U_t. Verification data V_t are available after that fitting interval and before commitment. The gate chooses $a_t = 1$ (commit) or $a_t = 0$ (retain), giving

$$f_t = \begin{cases} \widetilde{f}_t, & a_t = 1, \\ f_{t-1}, & a_t = 0. \end{cases} \tag{1}$$

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The initial state $M_{c,0}$ is assumed to be valid and may be obtained by any model-instantiation procedure. We focus on parameter-state maintenance after deployment and keep a_c fixed over the maintenance horizon. Repeated maintenance therefore produces the deployed-state sequence $M_{c,0} \to M_{c,1} \to \cdots \to M_{c,T}$.

Future Service data S_t become accessible only after the action and model identities are sealed. A Service interval may later mature into legally available Update information. This does not allow access to its labels when selecting its own preceding commitment.

Let $L(f; D)$ be mean squared prediction error on the valid target windows of interval D. Missing target labels are excluded by the registered temporal recipe; they are not reconstructed for the present evaluation. Forecast horizons use disjoint target roles and the frozen boundary purges. Model inputs can themselves be corrupted, while the underlying Service targets used for outcome auditing are held out from the corruption intervention. This controlled evaluation assumption is examined in Section 5.3.

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Original 3.2: Operational Data and Model-State Transition

3.2 Utility, Harm, and Adaptation

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At maintenance stage t, center c provides newly matured operational data $\mathcal{D}_{c,t}^{\mathrm{maint}}$. The maintenance data are characterized from two complementary perspectives:

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$$\mathcal{D}_{c,t}^{\mathrm{maint}} = \mathcal{D}_{c,t}^{\mathrm{ben}} \uplus \mathcal{D}_{c,t}^{\mathrm{cor}}, \qquad \mathcal{D}_{c,t}^{\mathrm{maint}} = \mathcal{D}_{c,t}^{\mathrm{upd}} \uplus \mathcal{D}_{c,t}^{\mathrm{ver}},$$

where $\uplus$ denotes disjoint union. The first partition describes observation reliability: $\mathcal{D}_{c,t}^{\mathrm{ben}}$ contains legitimate operating observations, whereas $\mathcal{D}_{c,t}^{\mathrm{cor}}$ contains unreliable observations caused by measurement faults, stale data, communication faults, or manipulation. Their membership is unknown to the maintenance algorithm. The second partition specifies algorithmic roles. The update subset $\mathcal{D}_{c,t}^{\mathrm{upd}}$ is used to generate a candidate state, whereas the chronologically later verification subset $\mathcal{D}_{c,t}^{\mathrm{ver}}$ is excluded from candidate adaptation and provides evidence for the deployment decision. Either subset may contain unreliable observations.

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For a prescribed candidate-adaptation operator $\mathcal{A}$, the candidate is

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$$\widetilde{\theta}_{c,t} = \mathcal{A}\Big(\theta_{c,t-1}, \mathcal{D}_{c,t}^{\mathrm{upd}}\Big), \qquad \widetilde{M}_{c,t} = (a_c, \widetilde{\theta}_{c,t}).$$

Candidate generation does not by itself modify the deployed model. Let $u_{c,t} \in \{0, 1\}$ denote the deployment decision, where $u_{c,t} = 1$ commits the candidate and $u_{c,t} = 0$ retains the current state. The deployed state after maintenance is

DELETED Original Equation 4

$$M_{c,t} = \begin{cases} \widetilde{M}_{c,t}, & u_{c,t} = 1, \\ M_{c,t-1}, & u_{c,t} = 0. \end{cases}$$

Thus, candidate adaptation and deployment commitment are distinct stages.

Define the retrospective Service loss difference $z_t = L(\widetilde{f}_t; S_t) - L(f_{t-1}; S_t)$. A harmful transition is an actual commitment with $z_t > 0$. For N maintenance opportunities, define harmful-commit frequency h, mean downside D, and commitment frequency C as

$$h = \frac{1}{N}\sum_t a_t \mathbf{1}[z_t > 0], \qquad D = \frac{1}{N}\sum_t a_t \max(z_t, 0), \qquad C = \frac{1}{N}\sum_t a_t. \tag{2}$$

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Original 3.3: Continual DT Model Maintenance Problem

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Let $\mathcal{L}(\mathcal{D}; M)$ denote the mean task loss of model state M on dataset $\mathcal{D}$. Let $\mathcal{D}^{\mathrm{srv}}_{c,t}$ denote the service interval following the stage-t deployment decision. These data are unavailable when $u_{c,t}$ is determined and are used only to evaluate the subsequent consequence. Define

DELETED Original Equation 5

$$\Delta_{c,t} = \mathcal{L}\big(\mathcal{D}^{\mathrm{srv}}_{c,t}; \widetilde{M}_{c,t}\big) - \mathcal{L}\big(\mathcal{D}^{\mathrm{srv}}_{c,t}; M_{c,t-1}\big).$$

A negative $\Delta_{c,t}$ indicates that the candidate would improve subsequent service performance relative to retaining the current state, whereas a positive value indicates degradation. A harmful state transition therefore occurs when the candidate is committed and $\Delta_{c,t} > 0$.

The denominator includes retained opportunities. These are not conditional error probabilities among committed candidates. Lifecycle utility is the mean deployed Service loss $L(f_t; S_t)$, aggregated equally over the specified stages, seeds, and centers. Main balanced MSE equally averages the clean and mixed streams; coherent-sticky is a separate risk stress and is not a third term in that utility average.

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For the prescribed candidate-adaptation operator $\mathcal{A}$, the ideal deployment-decision problem is

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$$\begin{aligned} \min_{\{u_{c,t}\}} \quad & \frac{1}{|C|T}\sum_{c\in C}\sum_{t=1}^{T} \mathcal{L}\big(\mathcal{D}^{\mathrm{srv}}_{c,t}; M_{c,t}\big) \\ \text{s.t.} \quad & \frac{1}{|C|T}\sum_{c\in C}\sum_{t=1}^{T} u_{c,t}\mathbf{1}[\Delta_{c,t} > 0] \le \rho, \\ & \frac{1}{|C|T}\sum_{c\in C}\sum_{t=1}^{T} u_{c,t}[\Delta_{c,t}]_+ \le \epsilon, \end{aligned}$$

where $|C|$ is the number of computing centers, $\mathbf{1}[\cdot]$ is the indicator function, $[z]_+ = \max(z, 0)$, $\rho \in [0, 1]$ is the tolerated harmful-transition frequency, and $\epsilon \ge 0$ is the tolerated average harmful-transition downside.

Source-domain empirical qualification requires $h \le 0.05$ and $D \le \eta$ in every required stream, where $\eta = 5.6036475532487204 \times 10^{-5}$ is frozen in the experiment. A method is active if it commits at least once in the clean/mixed evaluation. Qualification is an observed finite-sample condition, not a constraint enforced from inaccessible Service labels at runtime. An inactive Frozen policy can satisfy it trivially, so activity and utility are reported separately.

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Equation Original 6 defines an ideal lifecycle objective and risk criteria rather than an online guarantee. At maintenance time, $\mathcal{D}^{\mathrm{srv}}_{c,t}$ and hence $\Delta_{c,t}$ are unavailable. The practical problem is therefore to determine $u_{c,t}$ using only information available before the subsequent service interval, while maintaining lifecycle utility and limiting harmful deployed-state transitions.

Beneficial-candidate recall measures the fraction of audit-labelled opportunities with $z_t < 0$ that are committed. It uses each method's own causal parent/candidate chain. The supplementary tables retain both equal-seed/equal-center macro summaries and count-based micro denominators where provided. Neither recall nor commitment rate establishes cross-method equality of candidate opportunities. Window counts and stage counts are also not independent asset replications.

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Original 4: Proposed RAVM-DT Algorithm

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Original 4.1: Overview

4 Joint Evidence Gate

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RAVM-DT solves the decision problem in Section Original 3 by separating candidate adaptation from deployment commitment. At stage t, update data produce one candidate state, recent and bounded historical verification evidence compare that candidate with the current deployed state, and a selective decision either commits the candidate or retains the current state. Current verification data enter the historical buffer only after the decision. Figure Original 2 summarizes this workflow.

Figure 2 shows the causal maintenance ordering.

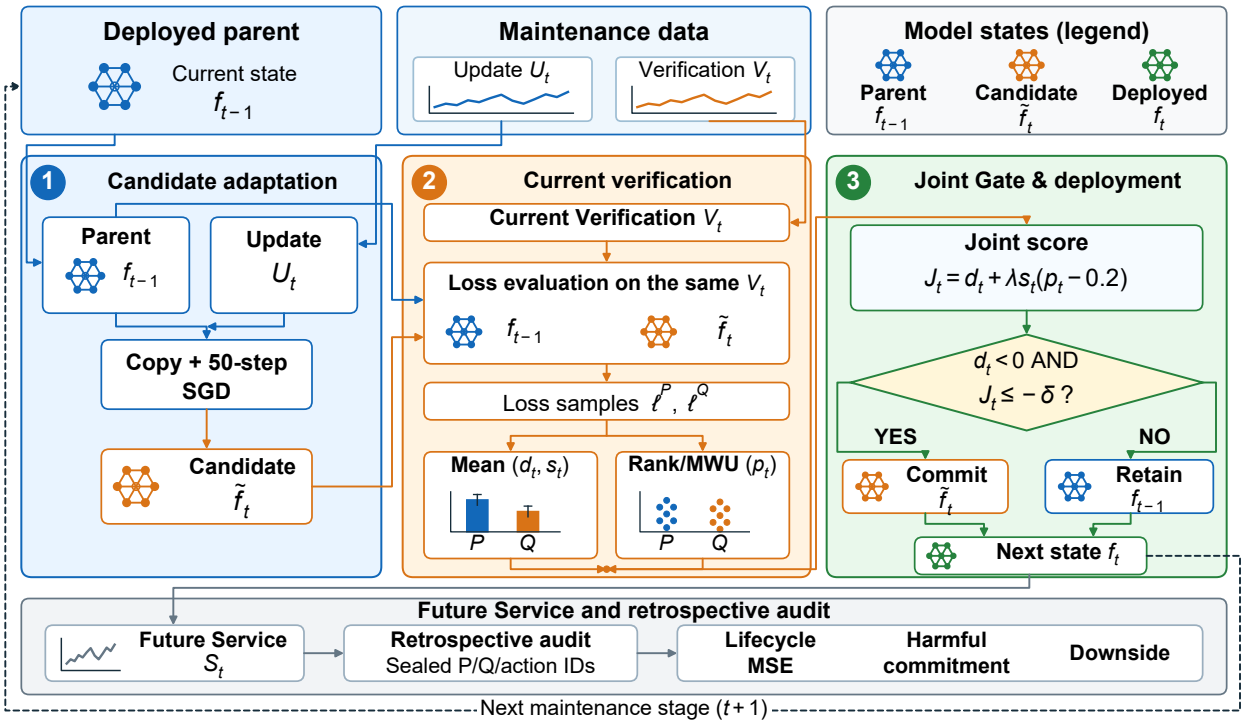


Figure 2: Joint Gate workflow: adaptation, current Verification, deployment and retrospective Service audit. P/Q identify the parent/candidate; green denotes the selected deployed state. All plotted marks are schematic. Model/action identities are sealed before Service access (Equation 4). AI-assisted artwork: OpenAI Codex (client 0.163.0-alpha.11); rendering: Matplotlib 3.10.0.

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Original 4.2: Candidate-State Adaptation

4.1 Current Verification Score

Let n_t be the number of valid windows in the current Verification interval. For each such window i, compute parent loss ℓ_i^P and candidate loss ℓ_i^Q. Define

$$d_t = \frac{1}{n_t}\sum_i(\ell_i^Q - \ell_i^P), \qquad s_t = \max\left(\frac{1}{n_t}\sum_i \ell_i^P, 10^{-12}\right). \tag{3}$$

The same windows supply both losses, so d_t is a paired loss difference. The one-sided rank output p_t is computed by `mannwhitneyu(new, old)` with `alternative='less'` and `method='asymptotic'`; it compares the two loss samples rather than their paired differences. Losses from six chronological Verification blocks are pooled for d_t and p_t. As discussed in Section 2, overlapping windows limit the probabilistic interpretation of p_t.

The joint score and action are

$$J_t = d_t + \lambda s_t(p_t - 0.2), \qquad a_t = \mathbf{1}[d_t < 0 \ \wedge \ J_t \le -\delta]. \tag{4}$$

The exact frozen values are $\lambda = 0.3$ and $\delta = 5.6036475532487204 \times 10^{-5}$. Inputs must be finite, the parent mean and parameters nonnegative, and $p_t \in [0,1]$ under the validated interface. Invalid evidence is an execution/audit failure, not an accepted deployment.

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At each stage, the candidate starts as an isolated copy of the current deployed state and is optimized only with update data. Let $\mathcal{B}_{c,t}^{(s)} \subseteq \mathcal{D}_{c,t}^{\mathrm{upd}}$ denote the update mini-batch used at step s. A gradient-based realization of the candidate-adaptation operator in Eq. Original 3 is

DELETED Original Equation 7

$$\begin{aligned}
\theta_{c,t}^{(0)} &= \theta_{c,t-1},\\
\theta_{c,t}^{(s+1)} &= \theta_{c,t}^{(s)} - \eta\nabla_\theta\mathcal{L}\big(\mathcal{B}_{c,t}^{(s)};(a_c,\theta_{c,t}^{(s)})\big), \quad s = 0,\dots,S-1,\\
\widetilde{\theta}_{c,t} &= \theta_{c,t}^{(S)}, \qquad \widetilde{M}_{c,t} = (a_c, \widetilde{\theta}_{c,t}),
\end{aligned}$$

where S is the stage-level adaptation budget and η the learning rate. The complete update subset defines the sampling pool, while $\mathcal{D}_{c,t}^{\mathrm{ver}}$ is excluded from parameter updates and the deployed state is unchanged until commitment.

For $p_t > 0.2$, the second term increases the required improvement; for $p_t < 0.2$, it discounts the score. The separate $d_t < 0$ condition excludes candidates whose current Verification mean loss is not lower than that of the deployed parent. At $p_t = 0.2$, the rule is the margin condition $d_t \le -\delta$. The parent-loss scale aligns the rank contribution with loss units within this task; it does not make the frozen margin invariant to normalization or establish risk calibration on another dataset.

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Original 4.3: Dual-Evidence Verification

4.2 Comparison with Neighboring Rules

RecentMean uses $d_t \le -\delta_R$; MWU-Mean uses $d_t \le -\delta_M$ and $p_t < 0.2$; DualMean uses $d_t \le -\delta_D$ and a nonpositive historical mean difference. Their frozen margins are $\delta_R = \delta_D = 4.20273566493654 \times 10^{-4}$ and $\delta_M = 2.8018237766243602 \times 10^{-5}$. Joint Gate therefore differs from a hard conjunction even when the current evidence is identical: a larger mean improvement can offset weaker rank evidence, while stronger rank evidence can support a smaller mean improvement. Its deployed lifecycle additionally differs because a commitment changes the next parent.

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Each verification collection is partitioned chronologically into B block units; target windows belong to one block, although their input contexts may overlap because of sliding-window construction. Let $\mathcal{E}^{\mathrm{rec}}_{c,t} = \mathcal{D}^{\mathrm{ver}}_{c,t}$ and let $\mathcal{E}^{\mathrm{hist}}_{c,t-1}$ denote blocks retained in a bounded historical buffer. Before the first online stage, $\mathcal{H}_{c,0}$ is initialized from target-specific pre-maintenance verification evidence. For $q \in \{\mathrm{rec}, \mathrm{hist}\}$ and chronological block $\mathcal{E}^{q}_{c,t,j}$, RAVM-DT evaluates the candidate and current state on the same block:

DELETED Original Equation 8

$$d^{q}_{c,t,j} = \mathcal{L}\big(\mathcal{E}^{q}_{c,t,j}; \widetilde{M}_{c,t}\big) - \mathcal{L}\big(\mathcal{E}^{q}_{c,t,j}; M_{c,t-1}\big).$$

Negative values favor the candidate. The paired differences are summarized by

DELETED Original Equation 9

$$\begin{aligned} m^{q}_{c,t} &= \mathrm{median}_j\, d^{q}_{c,t,j}, \\ r^{q}_{c,t} &= \mathrm{median}_j \big| d^{q}_{c,t,j} - m^{q}_{c,t} \big|, \\ U^{q}_{c,t} &= m^{q}_{c,t} + \kappa_q r^{q}_{c,t}, \qquad q \in \{\mathrm{rec}, \mathrm{hist}\}, \end{aligned}$$

where $\kappa_q \ge 0$ controls the penalty for disagreement among verification blocks. $U^{q}_{c,t}$ is a robust conservative evidence score, not a formal statistical confidence bound.

The original RAVM recent robust and bounded-history criteria and the RAVM-D2B implementation are retained as historical comparators. Joint Gate uses current evidence without a historical-loss condition. The common execution framework also computes historical scores for comparators, and that work remains included in measured maintenance time.

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Original 4.4: Verification-Guided State Transition

4.3 Reproducible Lifecycle Procedure

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Let $\delta_{\mathrm{imp}} \ge 0$ be the required recent improvement and $\delta_{\mathrm{ret}} \ge 0$ the tolerated historical degradation. The deployment decision is

DELETED Original Equation 10

$$u_{c,t} = \begin{cases} 1, & U^{\mathrm{rec}}_{c,t} \le -\delta_{\mathrm{imp}} \text{ and } U^{\mathrm{hist}}_{c,t} \le \delta_{\mathrm{ret}}, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, recent evidence requires adaptation benefit while historical evidence limits degradation of recently retained behavior. The historical buffer has capacity C_{hist}; after the stage-t decision, current verification blocks are appended and the oldest blocks are evicted if necessary. This ordering prevents current verification data from influencing their own decision through the historical branch and limits the effect of obsolete regimes.

Table 1 summarizes the causal lifecycle. The recorded model and data identities link each decision to the evidence available at commitment.

Table 1: Joint Gate lifecycle on each method's causal state chain.

Step	Operation
1	Load the authenticated source model and shared center/seed warm-up state f_0. Freeze parameters and temporal-role identities.
2	At stage t, obtain currently legal U_t, V_t; copy f_{t-1} and adapt the candidate on U_t only.
3	Evaluate both models on current Verification windows; compute d_t, s_t, p_t using Equation 3 and the frozen rank routine.
4	Apply Equation 4; persist candidate and parent identities, evidence identities, and the commitment before any access to S_t.
5	Deploy f_t using Equation 1. Only then evaluate Service utility, harm, and downside.
6	Advance temporal roles and causal state for the next maintenance opportunity.

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Algorithm 1 (RAVM-DT Maintenance for Center c at Stage t): 1. **Input:** $M_{c,t-1}$, $\mathcal{D}_{c,t}^{\mathrm{upd}}$, $\mathcal{D}_{c,t}^{\mathrm{ver}}$, historical buffer $\mathcal{H}_{c,t-1}$, and method parameters.
2. Adapt on $\mathcal{D}_{c,t}^{\mathrm{upd}}$ using Eq. Original 7 to obtain $\widetilde{M}_{c,t}$.
3. Form recent and historical evidence and compute Eqs. Original 8–Original 9.
4. Determine $u_{c,t}$ by Eq. Original 10 and update $M_{c,t}$ by Eq. Original 4.
5. **After the decision**, append current verification blocks to $\mathcal{H}_{c,t-1}$ and evict the oldest blocks if required.
6. **Return:** $M_{c,t}$ and the updated buffer $\mathcal{H}_{c,t}$.

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Original 4.5: Decision-Parameter Calibration

Computational reporting and assistive tools.

Deleted original block O059+O060+O061.

Let $\gamma \in \Gamma$ denote a candidate decision-parameter configuration. On calibration data disjoint from final evaluation, let $\widehat{\mathcal{R}}(\gamma)$, $\widehat{\mathcal{H}}(\gamma)$, and $\widehat{\mathcal{S}}(\gamma)$ denote empirical lifecycle loss, harmful-transition frequency, and harmful-transition downside. We select

DELETED Original Equation 11

$$\gamma^* \in \arg\min_{\gamma\in\Gamma} \widehat{\mathcal{R}}(\gamma) \quad \text{s.t.} \quad \widehat{\mathcal{H}}(\gamma) \le \rho, \qquad \widehat{\mathcal{S}}(\gamma) \le \epsilon.$$

The selected configuration is fixed before final evaluation. Equation Original 11 is an offline calibration rule; online decisions still use only information available before the subsequent service interval.

Experimental figures are rendered by the supplied Python/Matplotlib script from aggregate CSVs: it selects the stated method/stream groups, applies the displayed unit scales, and plots the recorded point estimates or bootstrap endpoints without refitting. OpenAI ChatGPT and Codex assisted software implementation/debugging, evidence organization, analysis workflows, figure-code preparation, literature organization and manuscript drafting/editing. Reported numerical results come from the executable workflows and their verification checks. Supplement S10 records recoverable tool/model versions, the AI-assisted preparation of Figure 2, and reproduction details.

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Original 5: Experimental Evaluation

5 Experiments

5.1 Experimental Setup

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Original 5.1.1: Benchmark and Lifecycle Protocol

5.1.1 Data, Temporal Roles, and Scope

Main uses 32 synthetic computing centers (63000–63031), three model seeds (7311–7313), three streams, twelve methods, and 24 opportunities per lifecycle: 3,456 trajectories and 82,944 method-stage records. The forecast target is a normalized latent workload, and the input consists of twelve noisy, delayed monitoring metrics and twelve observation masks. The latent process combines a center-specific logit level, three seasonal components, autoregressive fluctuations, and local bursts; a sigmoid maps it to $[0, 1]$. Metrics are nonlinear delayed transformations with heterogeneous sampling, schema loss, and missing observations. Each center contains 17,000 observations. With j denoting the zero-based observation index, four prescribed regimes add a logit shift q_j to the target and gain-scaled delayed shifts to the metrics: stationary has $q_j = 0$; abrupt changes to 0.8 at $j = 6520$; gradual ramps from zero at 6520 to 0.8 over 5,120 observations; recurring is 0.8 on $[5240, 9080)$ and $[12920, 17000)$, and zero otherwise. Regimes are balanced by center order, independently of the monitoring intervention. Supplement Sections S1–S3 give the full generator, role, and fault configurations. Synthetic data enable a known reference process, but they are not field observations. E1 and E2 use newly registered disjoint center sets with the same four regime categories. The supplied configuration sources and source-checkpoint index identify the executed generators and model assets.

For the workload protocol, warm-up occupies raw indices $[0, 800)$ and initial history $[800, 1000)$. At stage t, let $v_t = 1400 + (t - 1)640$. Update is $[v_t - 400, v_t)$, Verification $[v_t, v_t + 240)$, and Service $[v_t + 240, v_t + 640)$. Windows use 96 observations, horizon $H = 1$ or 4, and the frozen target-boundary purge. These workload coordinates are not imposed on the thermal recipes. UCI uses its registered chronological warm-up $[9000, 9300)$, initial history $[9300, 9500)$, and stage-specific intervals; ThermBuild retains its original observed-label coverage and 40 or 38 eligible stages. Supplement Section S5 and the supplied group configurations provide the exact interval lists, coverage exclusions, normalization, and target scales.

Alibaba Cluster Trace v2018 provides external machine utilization records [23]. E3 uses 40 previously exposed machines (20060–20099), three seeds, clean/mixed streams, and all twelve methods, yielding 2,880 trajectories. These machines are not a new blind field holdout; they test the frozen rule on an external workload source already encountered during development. The trace identity and processing recipe are retained. It is not a test of power-grid operation or autonomous control.

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Workload prediction is used as the maintenance task, following its role in proactive cloud/resource management [Original reference 28, 29, 30, 31]. Method development, decision calibration, and final evaluation use disjoint center pools; the final pool contains 30 centers, three fixed model seeds per center, and 24 chronological stages. For each target center, the initial deployed model is obtained by source pretraining followed by 50-step target warm-up adaptation on the first 800 raw observations. A disjoint 200-observation pre-maintenance interval initializes six historical verification blocks. Each stage then follows a raw-first causal split of 400 update, 240 verification, and 400 service observations; windows are formed separately within each role, giving 304/144/304 windows for $L = 96, H = 1$. The preceding service interval becomes the next stage's update interval only after its labels mature. The main predictor is a one-layer GRU-32; architecture/horizon generality is examined in Section Original 5.6.

The cross-task evaluation uses temperature targets T1, T2, and T3 from the UCI Appliances Energy Prediction dataset [24], containing 19,735 ten-minute observations from one house, and measured BSE1/BSE2 series from ThermBuild [25]. UCI rooms are correlated targets within one home, not independent buildings. ThermBuild's two experimental houses belong to one site; simulated cases are excluded. Missing target observations in the original thermal records are not filled to enlarge eligible coverage. Chronological origin blocks are formed before invalid-target exclusion. Normalized MSE and physical squared Celsius are reported separately, with no cross-task pooling.

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Original 5.1.2: Method Configuration and Calibration

5.1.2 Candidate Generators and Fair Comparison

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Method structure is fixed on the development pool and decision parameters are calibrated on a disjoint pool before final evaluation. Each verification subset is divided into six chronological blocks, and the 12-block history is updated by FIFO truncation only after the current decision. The decision search uses $\kappa_{\mathrm{rec}}, \kappa_{\mathrm{hist}} \in \{0, 0.5, 1.0, 1.5\}$; among configurations satisfying the absolute harmful-transition and downside criteria on mixed, persistent-bias, and replay calibration settings, lifecycle MSE is minimized with a deterministic tie-break. Table Original 1 reports the fixed configuration.

The twelve Main methods are Joint Gate, RecentMean, MWU-Mean, DualMean, Original RAVM, RAVM-D2B, Always-FT, ER, EWC, Huber-FT, ValidationCommit, and Frozen. Joint and the neighboring mean/rank gates share the same MSE candidate generator. ER and EWC use their native candidate-training policies and maintain separate causal state chains. Replay capacity is 24; EWC uses coefficient 10, Fisher decay 0.9 and 32 Fisher examples; the Huber loss parameter is 0.1. Legacy robust gates retain their original block configurations.

Candidates use 50 stochastic gradient descent (SGD) steps, learning rate 0.01, batch size $\min(64, n)$ with replacement, and gradient clipping at 1, where n is the number of eligible Update windows. All methods for a center/seed start from the same source checkpoint and 50-step target warm-up before their deployed states diverge. The default forecaster is a 32-hidden-unit gated recurrent unit (GRU) on 24 workload features, including observation masks. Architecture tests use a multilayer perceptron (MLP) with a flattened input and 64 ReLU units, or a two-layer causal temporal convolutional network (TCN) with 32 channels, kernel size 3, and dilations 1 and 2. Thermal models follow their 40- or 26-feature recipes (Supplement S2 and S5).

ValidationCommit uses the calibrated retain-all fallback, yielding the same deployed predictions as Frozen while still incurring candidate-generation cost. None of its five tested mean thresholds satisfied the mixed-stream harm limit: even the largest produced 123 harmful commits in 1,440 opportunities (8.54%). Supplement Table S2 gives the complete original calibration, separately from Main. RecentMean, MWU-Mean and DualMean provide active validation-based comparisons with independently selected margins.

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Original 5.1.3: Conditions, Baselines, Metrics, and Statistics

5.1.3 Development, Freezing, and Statistical Units

The saved DevCal search evaluates 16 Joint configurations: $\lambda \in \{0, 0.05, 0.15, 0.3\}$ and margins $\{1, 5, 10, 20\}\eta$. Eleven satisfy development risk eligibility; the selection pool contains nine active positive-weight configurations. The two eligible zero-weight settings are mean-only controls and are excluded from Joint candidate selection. Selection applies source risk qualification with an additional coherent-stream frequency requirement of at most 4.5%, then minimum clean/mixed balanced MSE; exact ties use measured maintenance cost, smaller weight, larger margin, and identifier. Neighboring mean gates are selected from their own recorded searches, preferring eligible active arms. Their distinct margins must not be interpreted as a controlled one-component intervention. The selection, protocol, and execution locks precede corrected Final training.

Development and evaluation are distinct: the stopped confirmation and the failed/corrected Smoke runs are development and exposure records, not additional independent confirmations of Main. Supplement S6 retains the stopping criteria, correction history and complete execution accounting.

Main reports every Joint-versus-baseline contrast. Its designated comparator is the lowest observed balanced-MSE baseline among all eleven alternatives that are empirically qualified and active; this yields DualMean. The resulting comparison is selected using Final outcomes, so its uncertainty does not account for comparator selection. Center-paired percentile bootstrap intervals use 4,000 resamples (seed 20261020 for Main; 20261022 for supplementary center/machine comparisons). All seeds, windows, and stages within an asset stay clustered. Intervals are unadjusted for multiplicity and are exploratory across contrasts. Thermal datasets lack enough independent buildings for population intervals; no stage- or seed-level pseudo-replication is substituted.

5.1.4 Registered Supplementary Coverage

The extended evaluations address five questions: monitoring severity (E1), architecture and forecast horizon (E2), external workload transfer (E3), measured temperature tasks (E4), and maintenance cost (E5). E1 uses eight methods, 24 new centers, three seeds and nine stresses (5,184 lifecycles). E2 uses six methods on 24 further centers in six GRU/MLP/TCN–horizon groups (864 lifecycles/group). E3 evaluates all twelve methods on the exposed Alibaba machines. E4 uses seven methods on three UCI targets (84 lifecycles each) and two ThermBuild targets (63 each). Together E1–E4 comprise 13,626 scientific lifecycles. Separately, E0 reconstructs Main summaries and replays 6,912 fixed Joint states without fitting; E5 comprises 36 timing repetitions. Supplement S4–S6 provides the full design and audit records.

5.1.5 Monitoring Interventions

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The four benign regimes apply a deterministic scalar workload shift: stationary remains unchanged, gradual ramps from 0 to 0.8, abrupt switches to 0.8 at the stage-9 verification boundary, and recurring alternates between shifted and unshifted intervals. The mixed condition overlays telemetry corruption with nominal fraction $\alpha = 0.10$ on each center's base regime. Each role segment is divided into ten temporal strata and $\lfloor n_s \alpha \rfloor$ positions per stratum are corrupted; mixed corruption cycles random noise, stage-persistent bias, bursts, and replay, while service data remain clean. Persistent-bias, replay, and additional α values are examined in Section Original 5.3.

E1 varies corrupted raw-observation fractions $\alpha \in \{0.05, 0.10, 0.20\}$. Mixed corruption affects Update and Verification, cycling through random noise, persistent bias, bounded bursts, and replay. Within ten chronological raw-observation strata, the implementation changes at most $\lfloor \alpha n_b \rfloor$ observations in each stratum b. Random perturbations use standard deviation 0.15, clipped to ± 0.2; bias uses signed 0.2 changes

on selected observed channels and targets, clipped to the normalized range. Replay draws only from a past warm-up segment. Persistent-bias and replay stress streams in E1 target Verification only, making the location of unreliable evidence explicit. Masks, seeds, and replay timestamps are common across methods.
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We compare RAVM-DT with Always-FT, experience replay (ER) [Original reference 20], elastic weight consolidation (EWC) [Original reference 16], Huber-loss fine-tuning (Huber-FT) [Original reference 25], ValidationCommit, and Frozen. Active methods share the 50-step adaptation budget; ER uses FIFO raw replay, EWC uses $\lambda = 10$ with Fisher EMA $\gamma = 0.9$, and Huber-FT uses $\delta = 0.1$. ValidationCommit uses the whole-verification mean candidate–current loss difference with a calibrated threshold; its calibration triggered the retain-all fallback, yielding zero final-pool commitments. Lifecycle MSE is the primary utility metric; transition safety is measured by harmful-transition frequency and average downside, with commit rate and beneficial-commit recall characterizing adaptation. Centers are the statistical units: seeds are averaged within center and paired 95% intervals use 4000 center-cluster bootstrap repetitions. For all methods, service data are excluded from candidate adaptation and deployment commitment.

Sensor traces document noise, spikes and abnormal constant readings [26]; networked estimation also encounters missing and delayed observations [27]. These mechanisms motivate controlled monitoring stresses. Main's coherent-sticky stream fixes one of twelve metric channels and a signed offset of 0.2 per center. It perturbs that channel and the observed target on 10% of each Update and Verification segment throughout stages 1–24, sampling positions without replacement for each center/stage/role. Masks are retained and Service is untouched. Unlike Mixed, this stream maintains a coherent mechanism rather than cycling mechanisms within ten strata. Supplement S3 gives the channel/sign mapping, seeds and task-specific UCI bias/dropout/stale and ThermBuild 2°C offset/one-hour dropout recipes.

The fractions, amplitudes and durations are stress-test settings, not estimates calibrated to the target DT installations. Supplement S9 separates native missingness, operational fault labels and paired-reference measurement errors. Published HVAC benchmarks include simulated sensor biases of ±2 and ±4 °C [28]; public AHU field records [29] and paired-reference measurements [30] supply additional numerical context. Their different units and reference conditions leave target-site calibration incomplete.
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Original 5.2: Overall Lifecycle Maintenance Performance

5.2 Experimental Results

5.2.1 Main Risk–Utility Comparison

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Figure Original 3 shows the central safety–utility–plasticity trade-off. Active always-update baselines achieve lower MSE, whereas RAVM-DT incurs about 6.8%–13.2% higher MSE but reduces harmful-transition frequency by about 85.5%–91.4% and harmful downside by 85.3%–94.9%. ValidationCommit makes no final-pool commitments and therefore follows Frozen; relative to this no-update reference, RAVM-DT lowers MSE by about 34.9% while committing roughly one quarter of candidate states. Thus, its safety gain is not obtained by disabling adaptation.

Table 2 and Figure 3 compare prediction utility under the stated empirical risk qualification.

Joint balanced MSE is 0.00318977355373 against DualMean's 0.00335236467765, a 4.8500% relative reduction. Joint improves in 29 of 32 paired centers; the center-paired MSE difference has 95% interval $[-0.000228546, -0.000103565]$. Its 19.0538% commitment rate across clean/mixed opportunities indicates active adaptation. The comparator-selection and interval conditions are defined in Section 5.1.3.

Joint's harmful commitments are 30/2,304 (1.3021%) in clean, 84/2,304 (3.6458%) in mixed, and 92/2,304 (3.9931%) in coherent-sticky. Corresponding mean downside values are 8.73992×10^{-7}, 9.48998×10^{-6}, and 1.07714×10^{-5}, all within the source qualification. DualMean, MWU-Mean, and RAVM-D2B are also active and qualified. The original historical RAVM exceeds the frequency limit in at least one stream.

Table 2: Main results for twelve methods. MSE/commitment balance clean and mixed; harm/downside are stream-wise maxima. Empirical qualification requires both limits in every stream.

Method	MSE $\times 10^3$	Max. harm (%)	Max. downside $\times 10^5$	Commit (%)	Qualified
Always-FT	2.9551	45.96	11.624	100.00	No
DualMean	3.3524	4.90	1.487	11.83	Yes
ER	3.0829	38.80	6.257	100.00	No
EWC	2.9546	45.96	11.393	100.00	No
Frozen	5.1816	0.00	0.000	0.00	Yes
Huber-FT	3.1226	40.19	5.383	100.00	No
Joint Gate	3.1898	3.99	1.077	19.05	Yes
MWU-Mean	3.3801	1.17	0.333	10.00	Yes
RAVM-D2B	3.3694	3.73	0.735	15.86	Yes
Original RAVM	3.3392	8.20	1.766	24.65	No
RecentMean	3.2139	5.47	1.817	16.19	No
ValidationCommit	5.1816	0.00	0.000	0.00	Yes

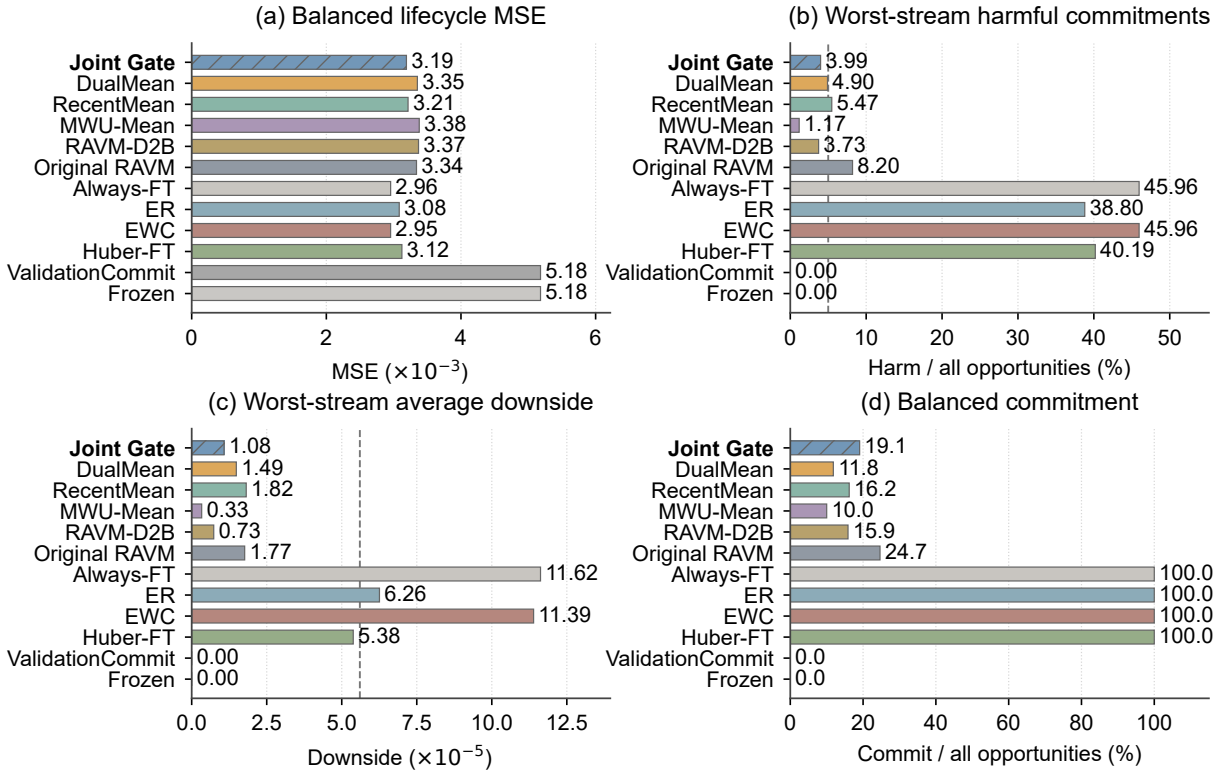


Figure 3: Main comparison (32 centers, three seeds): (a,d) clean/mixed balanced MSE and commitment; (b,c) worst-stream harm and downside. Rates use all opportunities. Bars are aggregates without intervals; dashed lines mark empirical qualification limits.

Always-FT, EWC, ER, and Huber-FT have lower raw balanced MSE than Joint, but exceed the required harm qualification. RecentMean's MSE is close to Joint's, and its paired interval $[-5.46006 \times 10^{-5}, 1.80747 \times 10^{-5}]$ crosses zero; its worst harm frequency is 5.4688%. These comparisons expose the prediction utility forgone to satisfy the specified risk preference. All eleven paired contrasts are reported in Section 5.2.3.

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Original 5.3: Robustness to Unreliable Monitoring Data

5.2.2 Monitoring Severity and Failure

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Across the nine tested corruption–severity combinations in Fig. Original 4, harmful-transition frequency remains below the 5% synthetic-domain calibration criterion. Mixed corruption produces the clearest utility/downside deterioration as α increases, while commit rates decline across all three corruption types. The safety metric is not monotone in nominal corruption fraction, so the result is interpreted as robustness within the evaluated range rather than a general severity law.

Figure 4 and Table 3 examine whether qualification persists as monitoring severity increases.

Table 3: Joint Gate under E1 stresses (1,728 opportunities/stream).

Monitoring stress	MSE $\times 10^3$	Harm / opportunities	Harm (%)	Commit (%)
mixed_a005	3.2961	35/1728	2.03	19.85
mixed_a010	3.3411	78/1728	4.51	21.35
mixed_a020	3.4212	147/1728	8.51	25.87
persistent_bias_a005	3.2953	19/1728	1.10	16.72
persistent_bias_a010	3.3394	14/1728	0.81	14.93
persistent_bias_a020	3.3966	9/1728	0.52	12.33
replay_a005	3.3036	15/1728	0.87	15.62
replay_a010	3.3239	8/1728	0.46	14.00
replay_a020	3.3963	5/1728	0.29	11.40

At 20% Mixed corruption, Joint makes 147 harmful commitments in 1,728 opportunities (8.5069%), exceeding the source frequency limit despite mean downside remaining below η. Replay and persistent Verification bias yield lower harm together with lower commitment. Mixed perturbs both candidate training and Verification, whereas the targeted stresses affect Verification only, so their differences combine fault mechanism and fault location.

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Original 5.4: Adaptation under Benign Operating Evolution

5.2.3 Benign Evolution and Same-State Behavior

Table 4 and Figure 5 examine adaptation to legitimate evolution under clean monitoring.

Table 4: Joint Gate under Main clean monitoring; beneficial-commit counts use audit-labelled candidates on Joint's own state chain.

Regime	Opportunities	Commit (%)	Harm (%)	Beneficial commits / candidates
abrupt	576	17.53	1.74	91/457
gradual	576	16.49	0.69	91/497
recurring	576	26.22	1.74	141/454
stationary	576	11.46	1.04	60/454

Joint remains active in all four regimes, with commitment rates from 11.46% in stationary centers to 26.22% in recurring centers. Beneficial-candidate counts also show missed opportunities. Each descriptive subgroup contains eight centers; recall is evaluated on each method's own candidate chain.

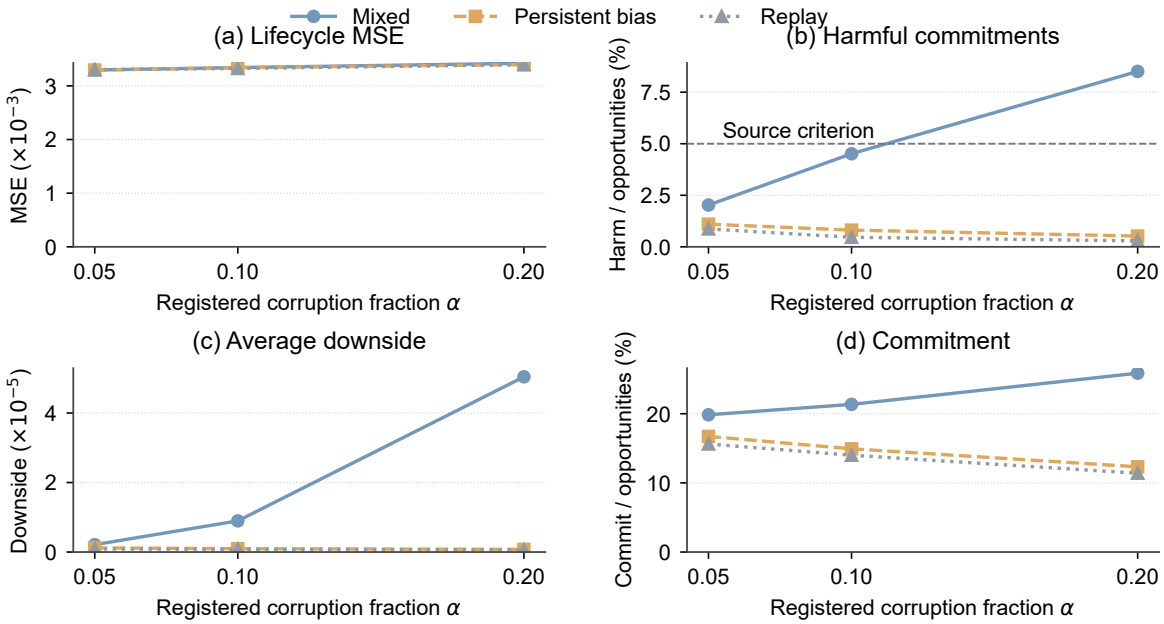


Figure 4: E1 monitoring stresses (24 centers, three seeds; 1,728 opportunities/stream). Panels: MSE, harm, downside and commitment. Points are aggregates without intervals; lines connect registered stress fractions. Dashed lines mark empirical limits.

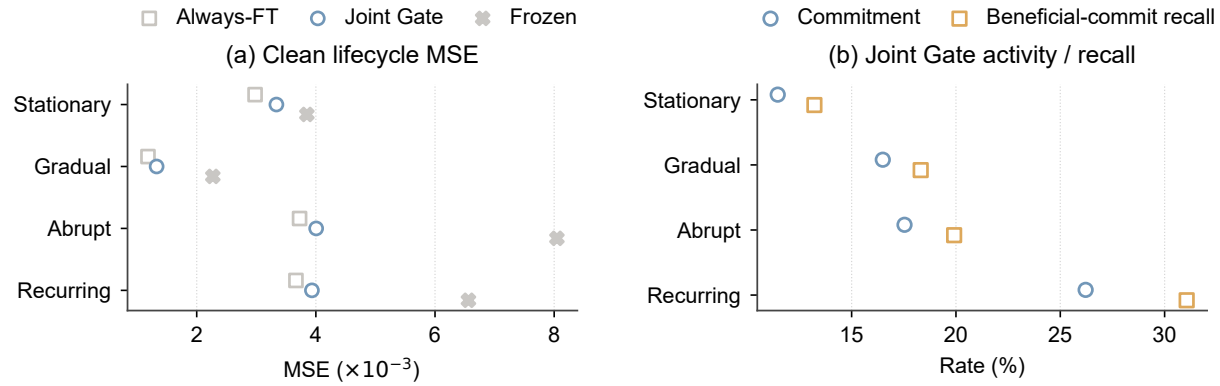


Figure 5: Main clean monitoring: (a) lifecycle MSE; (b) Joint commitment and beneficial-candidate recall. Each regime contains eight centers and 576 opportunities/method. Points are descriptive aggregates; recall uses method-specific beneficial candidates.

Same-state replay tests whether the rules make different decisions when model and Verification inputs are held fixed (Figure 6(b)). On 6,912 Joint parent/candidate/Verification states, Joint and DualMean disagree 971 times: 731 Joint-only and 240 Dual-only commitments. The corresponding disagreements are 699 against MWU-Mean (696 versus 3) and 758 against RecentMean (411 versus 347). Equation 4 reproduces all stored Joint actions. These are decision differences on Joint's state chain; component-level causal effects require separate interventions.

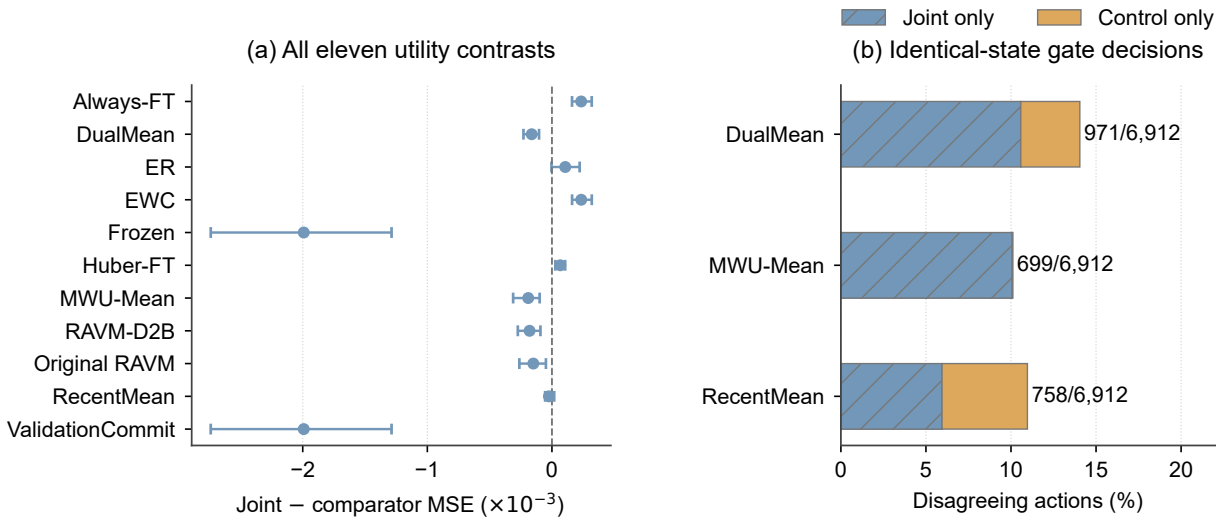


Figure 6: (a) All eleven Main Joint-minus-comparator MSE contrasts; negative favors Joint. Intervals: unadjusted 95% center-paired percentile bootstrap (32 centers, 4,000 draws). (b) Joint-only/control-only commitments on 6,912 identical states; labels count disagreements.

5.2.4 Architecture and Forecast Horizon

Architecture and forecast-horizon tests assess whether the source comparison extends beyond the default GRU/H1 (Table 5).

Table 5: E2 Joint-minus-DualMean balanced MSE and Joint mixed-stream harm. Intervals: unadjusted 95% center-paired bootstrap (24 centers, 4,000 draws).

Architecture / horizon	Δ MSE $\times 10^3$	95% interval $\times 10^3$	Mixed harm (%)	Qualified
GRU / H1	-0.1183	[-0.1797, -0.0470]	4.34	True
GRU / H4	-0.1949	[-0.2607, -0.1382]	5.44	False
MLP / H1	-0.1768	[-0.2794, -0.0933]	3.70	True
MLP / H4	-0.1325	[-0.2108, -0.0617]	5.50	False
TCN / H1	-0.0785	[-0.1322, -0.0263]	3.07	True
TCN / H4	-0.1931	[-0.3048, -0.0990]	5.73	False

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Always-FT remains the lowest-MSE reference, but RAVM-DT substantially outperforms Frozen under gradual, abrupt, and recurring drift. Its commit rate and beneficial-commit recall are higher under gradual, abrupt, and recurring drift than under stationary conditions, while harmful-transition frequency remains about 1.0%–2.6% across the four benign regimes. The method therefore retains selective plasticity at a measurable utility cost relative to unrestricted adaptation.

All six groups have negative Joint-minus-DualMean balanced-MSE differences with intervals excluding zero (Table 5; Figure 7(c)). However, the GRU, MLP, and TCN horizon-four groups exceed the mixed-stream empirical risk limit. The utility pattern against the fixed Main comparator therefore coexists with a horizon-dependent qualification failure.

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Original 5.5: Mechanism Analysis

5.2.5 Development Sensitivity

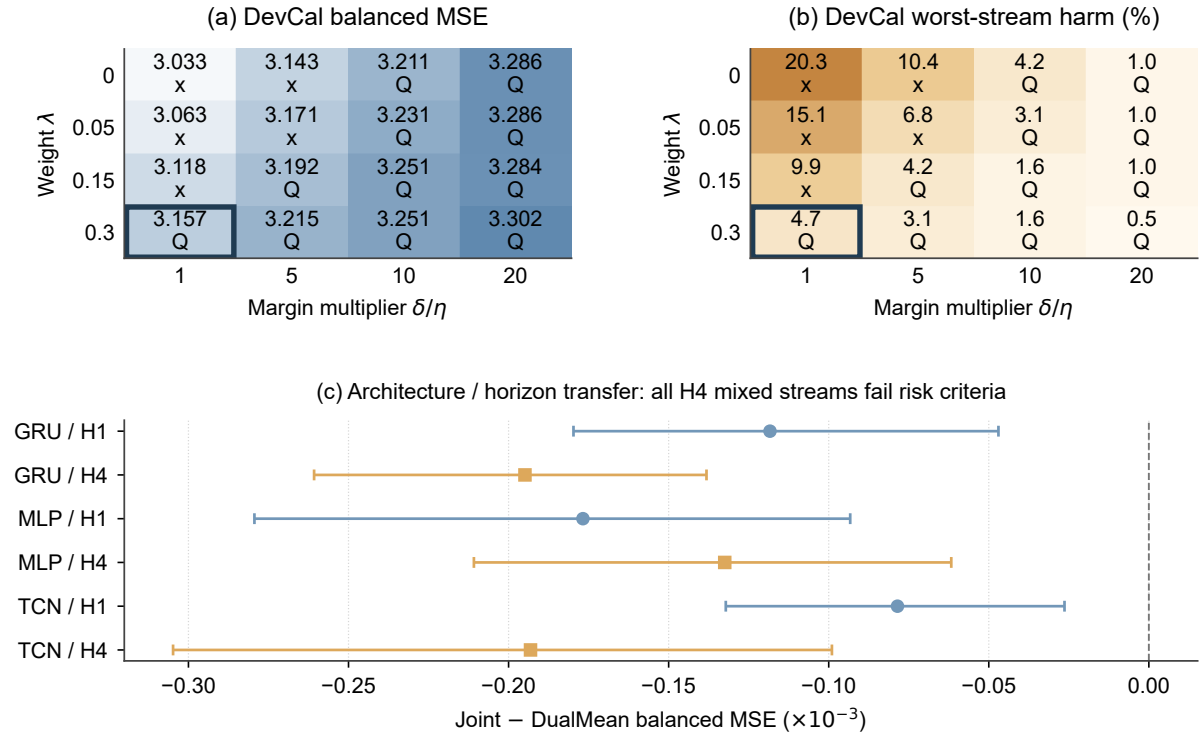


Figure 7: (a,b) Frozen DevCal grid: balanced MSE ($\times 10^{-3}$) and worst-stream harm (%). Q/x mark full development qualification/failure; outlines mark $\lambda = 0.3$, $\delta = \eta$. (c) Six E2 Joint-minus-DualMean contrasts; unadjusted 95% center-paired percentile intervals (24 centers, 4,000 draws).

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Figure Original 6(a) provides no evidence that the tested candidate-side complexity variants—blockwise construction or trimmed state-delta aggregation—improve the final maintenance outcome, supporting the simpler single-path candidate. Figure Original 6(b) shows that recent-only verification is more plastic but reaches about 12.6% harmful transitions under mixed corruption; adding history sharply reduces this exposure, while bounding the retained history recovers utility relative to a longer history. Finally, Fig. Original 6(c) shows that selective commitment strongly reduces harmful replacement at the cost of retaining more candidates. Together, these results support simple candidate adaptation followed by bounded temporal verification and selective deployment control.

Figure 7(a,b) shows how the development trade-off varies across the sixteen frozen configurations. Larger margins generally reduce activity, while the rank term changes the mean evidence required for commitment. A small margin with $\lambda = 0$ remains active but fails the empirical risk criterion. Eleven configurations meet development eligibility, and nine enter the active positive-weight selection pool. The selected $\lambda = 0.3$, margin-η setting was fixed before the extended evaluations.

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Original 5.6: Sensitivity, Generality, and Maintenance Cost

5.2.6 External Workload and Cross-Task Limits

The external workload evaluation tests transfer to 40 previously exposed Alibaba machines (Figure 8; Table 6).

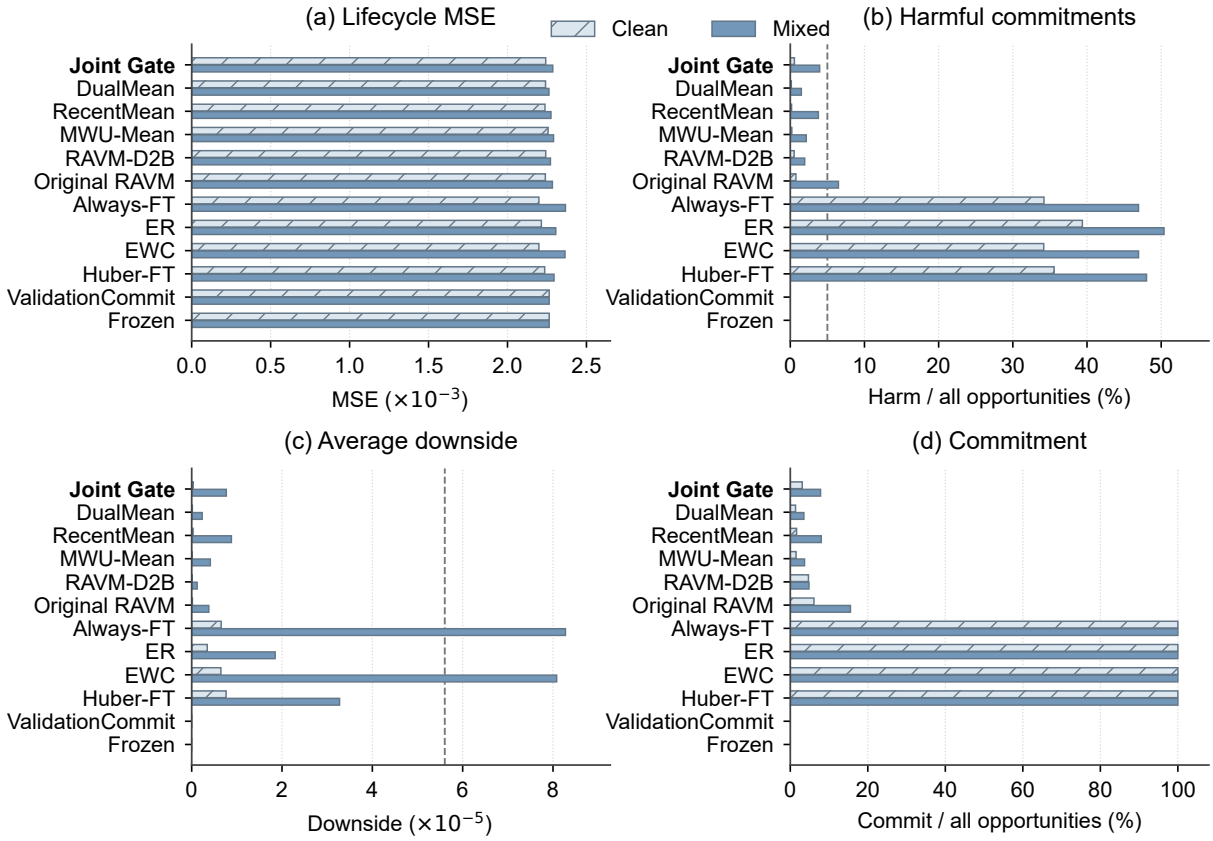


Figure 8: Alibaba evaluation on 40 previously exposed machines [23]: clean/mixed MSE, harm, downside and commitment (2,880 opportunities/method/stream). Bars are aggregates without intervals; dashed lines mark transferred empirical limits.

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The sensitivity grid in Fig. Original 7(a)–(b) exposes an explicit utility–risk trade-off; the fixed setting lies on the observed frontier but is not claimed to be universally optimal. Without architecture- or horizon-specific retuning, the expected qualitative ordering is retained in all 12 tested architecture–horizon–condition cells: RAVM-DT pays a positive MSE cost relative to Always-FT while reducing harmful-transition frequency. This supports consistency across the tested settings, not architecture-invariant performance.

Joint meets the transferred empirical risk criteria in both streams: harmful commits are 16/2,880 (0.556%) in clean and 114/2,880 (3.958%) in mixed, with mean downside 3.18493×10^{-7} and 7.66545×10^{-6}. DualMean also qualifies and has lower harmful frequency, downside and balanced MSE. The Joint-minus-DualMean balanced-MSE difference is $+1.26433 \times 10^{-5}$, with an unadjusted 95% machine-paired interval $[3.88690 \times 10^{-6}, 2.20413 \times 10^{-5}]$ ($n = 40$). Thus the empirical risk criterion transfers to these streams, but the source relative-utility advantage does not. Prior machine exposure limits independent confirmation. Supplement Table S5 gives all twelve methods, including Original RAVM's mixed harmful count of 187/2,880.

The temperature tasks test transfer beyond workload prediction (Table 7).

Table 6: Stream-specific Alibaba results for Joint and DualMean (2,880 opportunities/method/stream).

Method	Stream	MSE $\times 10^3$	Harm / opp.	Downside $\times 10^6$	Commit (%)
Joint Gate	clean	2.24203	16/2880	0.31849	3.090
Joint Gate	mixed	2.28668	114/2880	7.66545	7.812
DualMean	clean	2.24092	4/2880	0.14292	1.354
DualMean	mixed	2.26250	43/2880	2.32144	3.507

Table 7: Joint Gate on unmodified thermal streams. Failing-stream counts include registered interventions; physical-unit MSE uses each task's original target scale.

Task	Norm. MSE	MSE ($^\circ C^2$)	Harm / opp.	Harm (%)	Failing streams
uci T1	0.08729	0.2200	13/72	18.06	4/4
uci T2	0.14812	0.4025	14/72	19.44	4/4
uci T3	0.25042	0.5104	16/72	22.22	4/4
thermbuild BSE1	0.03039	0.0268	32/120	26.67	3/3
thermbuild BSE2	0.01068	0.0127	22/114	19.30	3/3

Every tested Joint thermal stream fails the transferred source risk qualification. Utility improvements in some comparisons therefore coexist with failure to satisfy the required deployment criterion. Three UCI room targets from one home and two ThermBuild houses at one site provide insufficient independent buildings for population-level intervals.

5.2.7 Measured Cost and Reproducibility

The timing comparison measures the maintenance cost of the complete 24-stage lifecycle (Table 8).

Table 8: E5 maintenance seconds per 24-stage lifecycle: mean [range] of three repeats on one center/seed/clean task. Service evaluation is excluded.

Method	Adapt.	Verify/gate	Persist.	Other	Maintenance mean [range]	Service
Always-FT	5.47	0.04	0.83	4.06	10.39 [10.15, 10.80]	0.48
DualMean	5.37	1.68	0.11	4.54	11.70 [11.47, 11.92]	0.89
ER	4.80	0.03	0.61	3.31	8.75 [7.52, 10.66]	0.44
EWC	6.54	0.03	0.58	4.09	11.25 [10.71, 11.77]	0.45
Frozen	0.00	0.03	0.27	1.91	2.20 [2.00, 2.39]	0.68
Huber-FT	5.24	0.03	0.76	3.33	9.36 [8.62, 10.14]	0.46
Joint Gate	7.74	1.56	0.47	4.51	14.28 [13.64, 14.89]	1.05
MWU-Mean	5.77	1.64	0.61	4.79	12.81 [12.03, 13.81]	0.97
RAVM-D2B	6.16	1.74	0.28	7.67	15.85 [14.35, 16.74]	1.42
Original RAVM	5.76	1.71	0.46	3.90	11.84 [10.44, 13.22]	0.90
RecentMean	5.48	1.55	0.34	4.64	12.02 [11.63, 12.39]	1.05
ValidationCommit	5.22	0.21	0.70	3.14	9.26 [7.94, 10.35]	0.47

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Table Original 2 shows that RAVM-DT requires about 0.185 s per maintenance stage, 55.7 MB peak GPU memory, and a 155.5 kB persistent buffer on the evaluated platform. Timing covers update construction, adaptation, verification, commitment, and maintenance bookkeeping, but excludes subsequent service evaluation and file I/O. Its runtime is of the same order as the adaptive baselines; these measurements characterize the implementation rather than a general latency guarantee.

Joint averages 14.28 s versus DualMean's 11.70 s across three repetitions. Maintenance includes adaptation, Verification/gating, model persistence, sealing/bookkeeping and baseline Fisher work before first Service access; shared warm-up and subsequent Service evaluation are excluded. ValidationCommit

incurs candidate-generation cost despite retaining every parent. These measurements use one ready lifecycle at a time per repetition on a shared host; the earlier parallel Main totals have a different timing context.

The host has 20 logical CPUs, approximately 16 GB system RAM, and an NVIDIA GeForce RTX 3060 Laptop GPU with 6 GB device memory. Supplement S6 and S10 specify the software environments, timing components and resource observations. Timing repetitions are engineering measurements rather than additional scientific samples.

Reproduction materials provide two complementary entries: read-only reconstruction from the supplied aggregate CSVs, and a bounded executable synthetic subset in `training_bundle`. The latter includes the required inputs, source/initial-model checkpoints, original code and relative-path commands for sixteen previously accepted fixed-input cases and one source-fitting-to-lifecycle case. The broader historical acceptance covered nineteen lifecycles, including measured-domain cases whose input/checkpoint bytes are not in this submission. Supplement S8–S10 and `08_Reproduction/README.md` distinguish the supplied paths, provider acquisition steps, stage-specific environments and completed validation. Full-matrix, all-domain source-path and independent external reproduction remain outside that acceptance.

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Original 5.7: External Production-Trace Evaluation

5.3 Discussion and Limitations

Joint Gate addresses a deployment trade-off: the specified empirical risk preference excludes some methods with lower raw MSE, while the active gate retains adaptation to legitimate change. The useful choice depends on the application's loss tolerance and outcome quality. Same-state disagreement establishes a behavioral distinction from neighboring gates; it does not assign the lifecycle gain to an individual score component.

The stress and transfer results show that parent-loss scaling alone does not make the selected margin or utility advantage portable. New domains require their own development protocol and independent evaluation. Reliable retrospective outcomes are equally important: delayed, biased or missing Service references can distort risk estimates and later maintenance feedback even when temporal access is correctly controlled.

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We finally evaluate the fixed configuration on the reconstructed 40-machine Alibaba Cluster Trace split [Original reference 32], without target-domain recalibration. For each selected machine, the first 17,000 valid timestamp-sorted observations are retained without interpolation; seven telemetry values are represented with value/mask channels and processed by the same causal lifecycle protocol. Under clean workloads, RAVM-DT is close to the active baselines in MSE while showing much lower harmful-transition exposure. Under mixed corruption, it remains selectively active and strongly reduces harmful transitions/downside relative to active baselines, but its harmful-transition frequency reaches 6.49%, above the 5% synthetic-domain calibration criterion, and its MSE is slightly higher than Frozen. The result therefore supports qualitative transfer of selective state-transition control, while showing that the exact risk operating point can be domain dependent.

Statistical replication is at the asset level: three workload seeds measure within-asset training variation, while the UCI rooms and ThermBuild houses provide limited independent replication. Bootstrap intervals omit multiplicity and selected-comparator uncertainty. The evaluated settings do not cover adversarial monitoring, all fault durations, control feedback or safety-critical intervention. The rank statistic under serial dependence and the selected thresholds provide empirical qualification rather than a formal guarantee of future risk.

6 Conclusion

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This paper presented RAVM-DT for continual maintenance of deployed DT models under evolving and potentially unreliable monitoring data. By separating candidate adaptation from deployment commitment, RAVM-DT uses paired recent and bounded historical verification evidence to decide whether an adapted candidate should replace the current state. Across the five primary lifecycle conditions, it reduces harmful-transition frequency by about 85.5%–91.4% and harmful downside by 85.3%–94.9% relative to active adaptation baselines, at a 6.8%–13.2% MSE increase; relative to Frozen, it lowers MSE by about 34.9% while retaining nonzero adaptation. Mechanism, benign-drift, and architecture/horizon results support the intended selective-transition behavior rather than a retain-all policy. On Alibaba workloads, the same qualitative safety control transfers, but the 6.49% mixed-condition harmful-transition frequency exceeds the 5% synthetic calibration criterion, indicating domain-dependent risk calibration. Current evidence is empirical and limited to workload prediction on one controlled benchmark and one external trace; future work will study calibration under domain shift, additional DT tasks/platforms, and mechanisms that improve adaptation without sacrificing transition safety.

Current mean improvement and rank evidence can be combined to make deployment decisions under unreliable monitoring. Joint Gate implements this trade-off with a negative-mean requirement and a causally ordered evaluation interface. In Main, it reduces balanced lifecycle MSE by 4.85% relative to the selected empirically qualified active DualMean baseline while retaining nonzero adaptation and meeting the observed risk criteria. Lower unrestricted MSE in other methods, the inconclusive RecentMean contrast, monitoring/horizon failures and the external-task results delimit this benefit. The evidence supports the specified predictive-model maintenance setting, with domain-specific validation needed for transfer.

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Original Author Contributions: The authors confirm contribution to the paper as follows: conceptualization, Jianyong Wang; methodology, Jianyong Wang; software, Jianyong Wang; validation, Jianyong Wang; formal analysis, Jianyong Wang; investigation, Jianyong Wang; data curation, Jianyong Wang; visualization, Jianyong Wang; writing—original draft preparation, Jianyong Wang; writing—review and editing, Jie Liu, Yubo Dong, Tengyu Gai and Fengbiao Zan. All authors reviewed and approved the final version of the manuscript.

Author Contributions: Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation, Jianyong Wang; review and editing, Jie Liu, Yubo Dong, Tengyu Gai and Fengbiao Zan.

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Original Data Availability: The implementation code, experimental configurations, and processed results supporting the findings of this study are available from the corresponding author upon reasonable request. The Alibaba Cluster Trace used for external evaluation is a publicly available third-party dataset and should be obtained from the original provider; it is not redistributed with this article.

Availability of Data and Materials: The Supplement and accompanying reproduction attachment provide aggregate CSVs, reconstruction scripts, configurations, recipe sources and identity indices. A bounded synthetic `training_bundle` supplies five group configurations with required processed inputs, source/initial-model checkpoints, original code and relative-path entries. Measured-domain training-input/checkpoint bytes, raw archives and intermediate candidate stores are not included. Alibaba, UCI and ThermBuild data must be obtained from the cited providers under their terms; UCI uses CC BY 4.0, ThermBuild CC BY-SA 4.0, and Alibaba research/study terms apply. Included derivatives retain attribution and source-specific conditions. Supplement S8–S10 gives acquisition instructions and the validated execution scope. No public repository deposit is claimed.

Ethics Approval: Not applicable. This study does not involve human participants or animals.

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Conflicts of Interest: The authors declare no conflicts of interest. The funder had no role in study design, collection, analysis or interpretation, manuscript preparation, or the decision to publish.

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Deleted Original Content

Historical objects below belong to the original submitted RAVM study; they are not additional results of the current Joint Gate. Original prose is already reproduced in the main comparison and is not repeated here.

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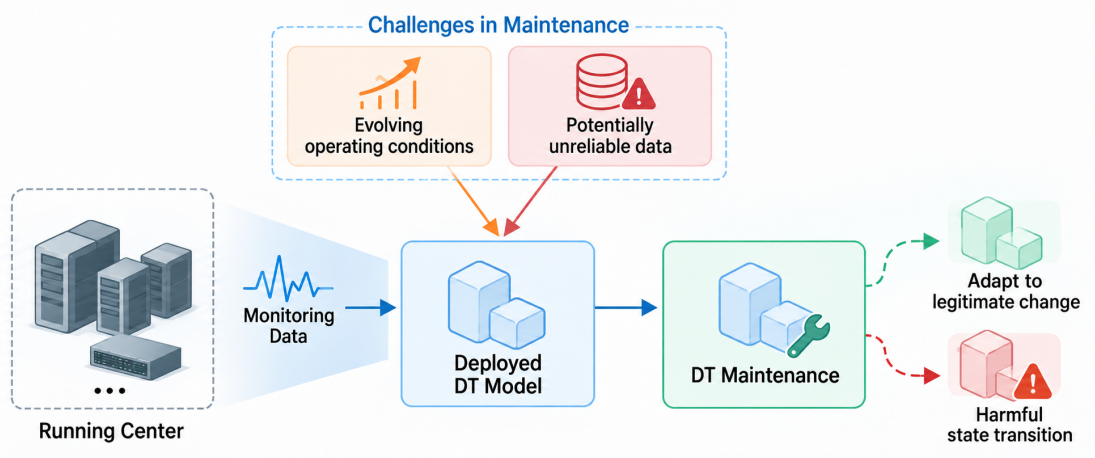
Verification-Guided Continual Maintenance of Digital Twin Models Under Evolving and Potentially Unreliable Monitoring Data

DELETED Original abstract

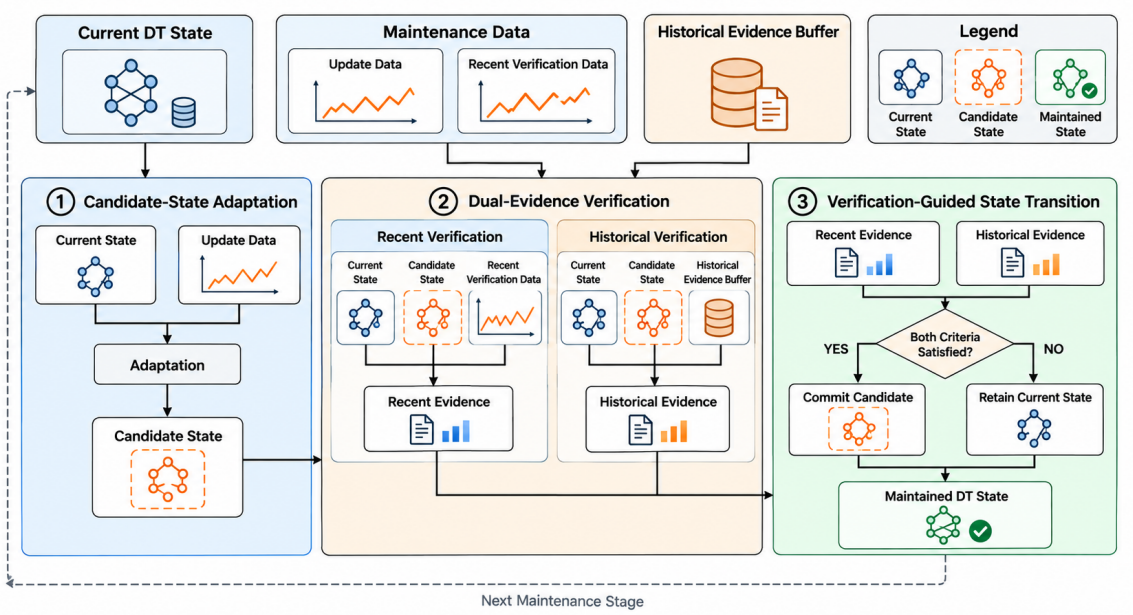
Digital twin (DT) models deployed in computing infrastructures require continual maintenance as operating conditions evolve and new monitoring data become available. However, newly accumulated monitoring data may reflect legitimate operating changes or unreliable observations, so directly updating the deployed model can improve adaptation but may also introduce harmful model-state transitions. Existing DT updating and continual-learning methods mainly address how models are adapted, leaving insufficiently controlled whether an adapted candidate should actually replace the current deployed state when maintenance evidence itself may be unreliable. To address this problem, we propose RAVM-DT, a verification-guided continual maintenance algorithm that explicitly separates candidate adaptation from deployment commitment. At each maintenance stage, RAVM-DT adapts the current model using update data, compares the resulting candidate with the deployed state using paired recent and bounded historical verification evidence, and commits the candidate only when the evidence supports the transition; otherwise, the current state is retained. Across the five primary lifecycle conditions, RAVM-DT reduces harmful-transition frequency by approximately 85.5%–91.4% and harmful-transition downside by 85.3%–94.9% relative to active continual-adaptation baselines, at a 6.8%–13.2% increase in lifecycle MSE. Compared with the no-maintenance reference, it reduces lifecycle MSE by approximately 34.9% while retaining nonzero adaptation. The selective-transition behavior remains qualitatively consistent across corruption severities, three model architectures, two prediction horizons, and an external production workload trace.

DELETED Original keyword

Digital twin; continual maintenance; model-state transition; verification; unreliable monitoring data; continual learning



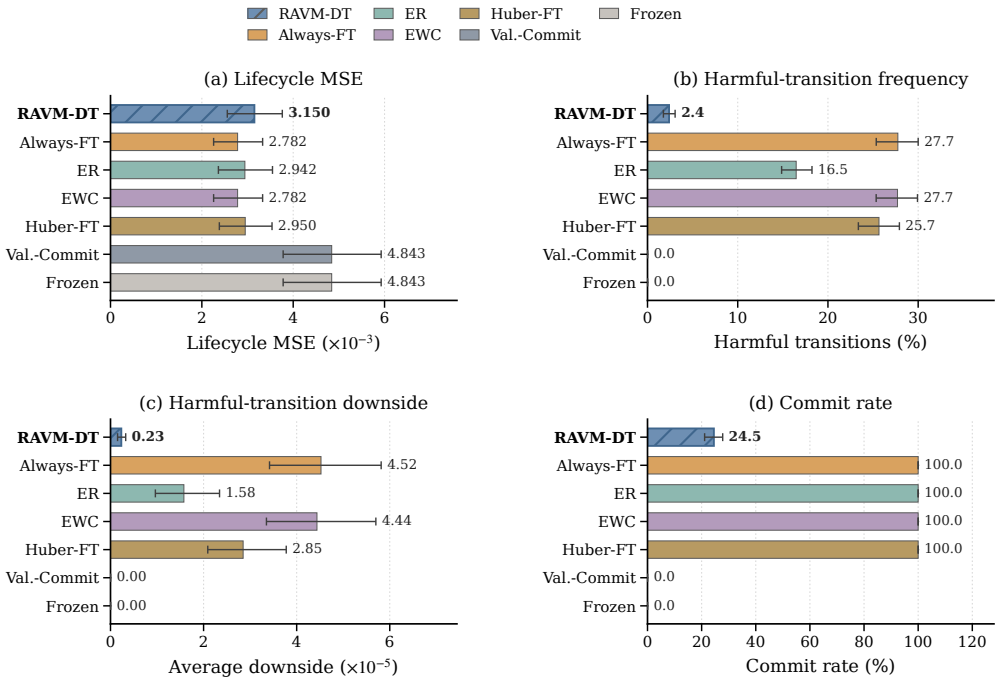
DELETED Original Figure 1: Problem setting for continual DT maintenance under evolving operating conditions and potentially unreliable monitoring data.



DELETED Original Figure 2: Overview of RAVM-DT. A single candidate state is adapted from the current DT state using update data, evaluated through recent and bounded historical evidence, and committed only when both criteria support replacement; otherwise, the current state is retained as the maintained DT state.

DELETED Original Table 1: Main Experimental Configuration

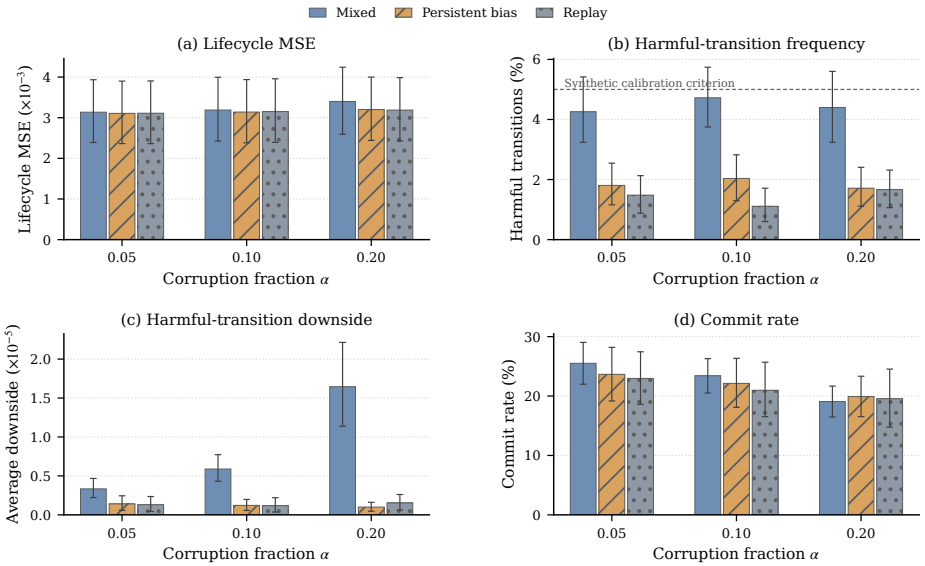
Category	Item	Setting
Lifecycle	Centers / seeds / stages	30 / 3 / 24
Predictor	Architecture / input / horizon	GRU-32 / $L = 96$ / $H = 1$
Adaptation	Optimizer / loss / updates	SGD / MSE / 50
	Learning rate / clipping	0.01 / 1.0
	Batch / sampling / weight decay	64 / replacement / 0
Verification	Blocks per stage / history capacity	6 / 12 blocks
	$\kappa_{\mathrm{rec}}, \kappa_{\mathrm{hist}}$	0, 0.5
Commitment	$\delta_{\mathrm{imp}}, \delta_{\mathrm{ret}}$	2.8018×10^{-5}, 0
Risk criteria	ρ, ϵ	0.05, 5.6036×10^{-5}
Statistics	Unit / bootstrap	Center / 4000 repetitions
	Model / attack / bootstrap seeds	7311–7313 / 9417 / 2904
Hardware	GPU	RTX 3060 Laptop GPU



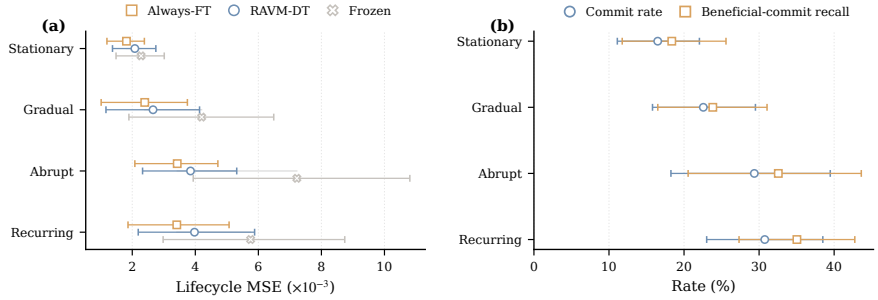
DELETED Original Figure 3: Overall lifecycle performance over the five primary conditions: (a) MSE, (b) harmful-transition frequency, (c) average harmful downside, and (d) commit rate. Error bars denote 95% center-cluster bootstrap intervals.

DELETED Original Table 2: Measured Maintenance Cost in the Final Benchmark

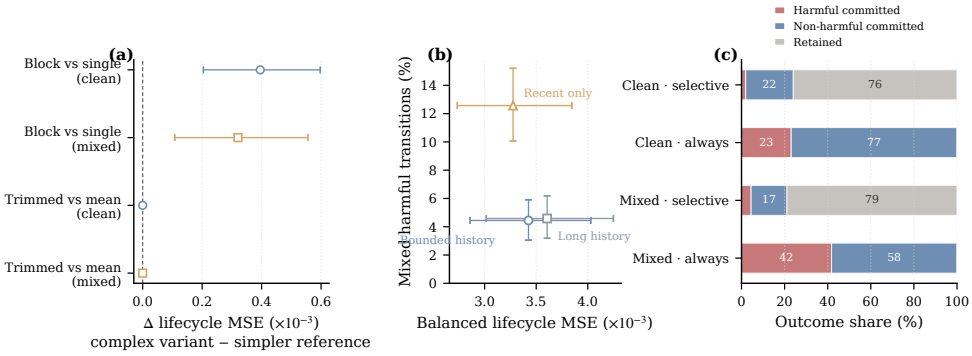
Method	Runtime/stage (s)	Peak GPU (MB)	Buffer (kB)
RAVM-DT	0.185	55.7	155.5
Always-FT	0.250	55.6	0
ER	0.262	98.3	302.4
EWC	0.504	58.3	44.8
Huber-FT	0.250	55.6	0
ValidationCommit	0.252	67.7	0
Frozen	0.004	25.5	0



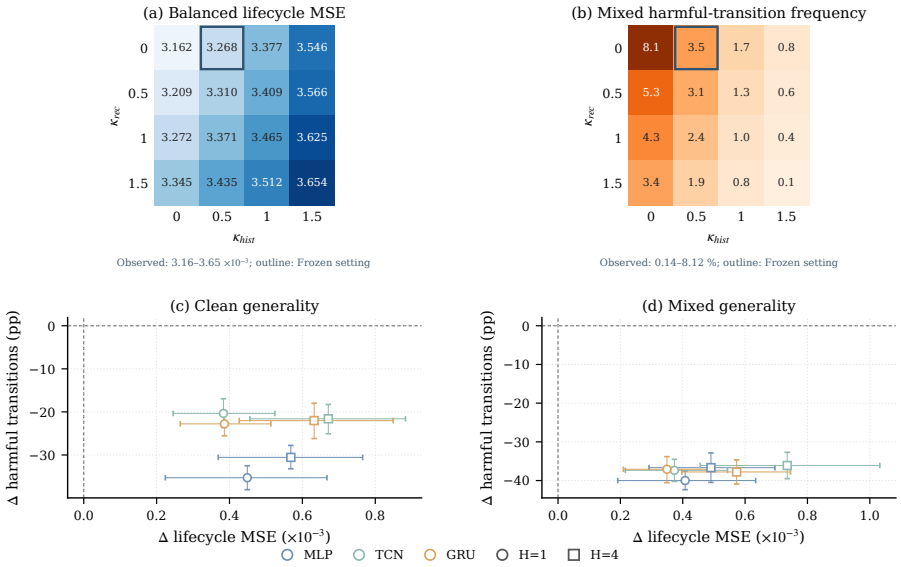
DELETED Original Figure 4: Robustness across corruption severities. Mixed corruption affects update and verification data; persistent bias and replay target verification evidence. Panels report MSE, harmful-transition frequency, harmful downside, and commit rate for $\alpha \in \{0.05, 0.10, 0.20\}$.



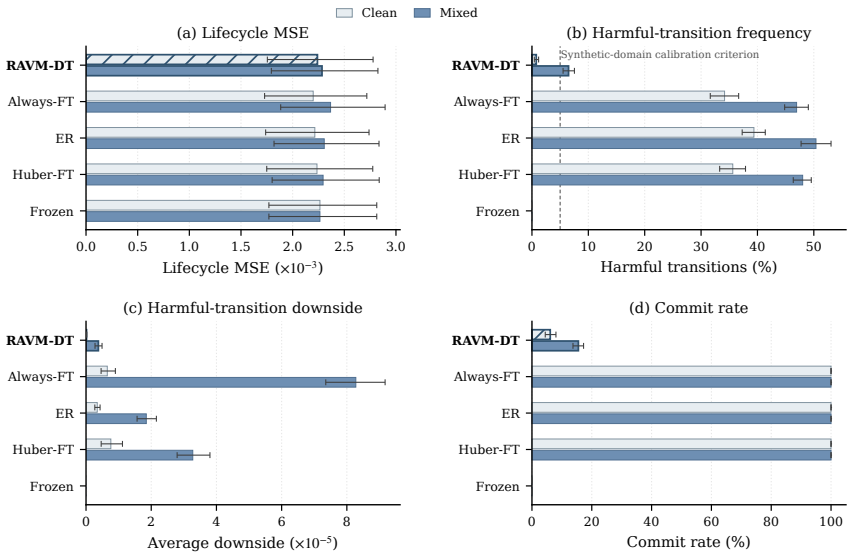
DELETED Original Figure 5: Benign operating evolution: (a) lifecycle MSE for RAVM-DT, Always-FT, and Frozen; (b) RAVM-DT commit rate and beneficial-commit recall. Error bars denote 95% center-cluster bootstrap intervals.



DELETED Original Figure 6: Mechanism analysis on the development pool: (a) paired MSE effects of additional candidate-side complexity, (b) utility–risk trade-off of recent-only, long-history, and bounded-history verification, and (c) deployment outcomes for selective and always-commit variants.



DELETED Original Figure 7: Sensitivity and generality of RAVM-DT. Panels (a)–(b) show the predefined $\kappa_{\text{rec}} \times \kappa_{\text{hist}}$ grid; the outlined cell is the fixed setting $(0, 0.5)$. Panels (c)–(d) show clean and mixed effects, computed as RAVM-DT minus Always-FT, across MLP, TCN, and GRU with $H \in \{1, 4\}$.



DELETED Original Figure 8: External Alibaba-trace evaluation under clean and mixed corruption. Panels report (a) lifecycle MSE, (b) harmful-transition frequency, (c) harmful downside, and (d) commit rate. The 5% line in (b) is the synthetic-domain calibration criterion.

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